

To Store Commands, Maintain History Buffer, Handle Capacity Limits, Retrieve Entries, Display History, Validate Consistency

Code:

```
geya@LenovoL480: ~/OS/Skil  ×  +  v
GNU nano 8.7.1 task1.c
#include <stdio.h>
#include <string.h>

#define MAX 5
#define SIZE 100

int main() {
    char history[MAX][SIZE];
    int count = 0;
    char command[SIZE];

    while (1) {
        printf("Enter command (type history or exit): ");
        fgets(command, SIZE, stdin);

        command[strcspn(command, "\n")] = '\0';

        if (strcmp(command, "exit") == 0)
            break;

        if (strcmp(command, "history") == 0) {
            printf("\nCommand History:\n");

            for (int i = 0; i < count; i++)
                printf("%d: %s\n", i + 1, history[i]);
        }
    }
}
```

```

        continue;
    }

    /* Handle capacity limit */
    if (count < MAX) {
        strcpy(history[count], command);
        count++;
    }
    else {
        /* Remove oldest command */
        for (int i = 0; i < MAX - 1; i++)
            strcpy(history[i], history[i + 1]);

        strcpy(history[MAX - 1], command);
    }

    printf("Command stored.\n");
}

return 0;
}

```

Output:

```

geya@LenovoL480:~/OS/Skill$ ./task1
Enter command (type history or exit): ls
Command stored.
Enter command (type history or exit): pwd
Command stored.
Enter command (type history or exit): history

Command History:
1:  ls
2:  pwd
Enter command (type history or exit): exit
geya@LenovoL480:~/OS/Skill$ |

```